

Overview of IPv4 format:

An IPv4 address has 32 bits, divided into 4 sections of 8 bits each. A typical IPv4 address is of the format as follows.

XXXXXXXX.XXXXXXXXXX.XXXXXXXXXX.XXXXXXXXXX

- Where each x represents either a **1** or **0**. Since it is not easy to remember the binary notation of 32 bits, it is converted into a human-readable decimal form before it is displayed to us e.g. 192.168.4.3.
- If we calculate the total number of devices that can be assigned a unique IPv4 address, the number comes to 2^{32} , which is 4,294,967,296 addresses. Since every device connected to the internet—such as computers, smartphones, servers, IoT devices, and devices in private or intranet networks—requires a unique IP address, the available pool of IPv4 addresses gets exhausted very quickly. This shortage of addresses is known as IPv4 address exhaustion and is one of the major reasons for the development of IPv6.
- The fact that [Réseaux IP Européens Network Coordination Centre](#) (an [RIR](#) that is responsible for the allocation and registration of IP addresses in Europe, Central Asia, Russia, and West Asia) has announced that it has fully run out of IPv4 addresses as of 25 November 2019, shows that it's a real problem. RIPE NCC is currently scraping IPv4 addresses from the organization that have gone out of business or from networks that return addresses they no longer need and is encouraging the shift from IPv4 to IPv6.
- Since the development of IPv4 addresses in 1984 by DARPA, it became clear in the first decade that there is a risk of IPv4 addresses running out, also known as **IPv4 exhaustion** i.e., the number of available IPv4 addresses being less than the number of devices placing requests for information. Attributing to the growth of personal computers and smart devices like speakers, T.V.'s, and watches, we began running out of the available IP addresses more quickly than it was first anticipated. An alternative had to be looked for.

Need for IPv6

IPv6 was developed to overcome the limitations of IPv4 and to support the growing number of internet users and devices. The major needs for IPv6 are as follows:

1. **Address Exhaustion Problem**
IPv4 uses 32-bit addressing, which provides about 4.3 billion unique addresses. Due to the rapid growth of computers, smartphones, IoT devices, and internet users, the available IPv4 addresses are almost exhausted. IPv6 uses 128-bit addressing, which provides a very large number of unique addresses.
2. **Better Security**
IPv6 has built-in support for security features such as IPSec, which provides authentication and encryption for secure communication.
3. **Efficient Routing**
IPv6 simplifies routing by using a hierarchical addressing structure, reducing the size of routing tables and improving network performance.
4. **Simplified Header Format**
IPv6 has a simpler header compared to IPv4, which allows faster packet processing and efficient data transmission.

5. Auto Configuration

IPv6 supports Stateless Address Auto Configuration (SLAAC), allowing devices to automatically configure their own IP addresses without requiring DHCP.

6. Improved Quality of Service (QoS)

IPv6 supports better handling of real-time applications such as video streaming, online gaming, and voice communication.

7. Support for Modern Networks

IPv6 is designed to support modern technologies such as IoT, cloud computing, and mobile networks more efficiently.

Therefore, IPv6 is essential for the future growth, scalability, and security of the Internet.

Overview of IPv6 format:

An IPv6 is a 128-bit address represented in 8 groups of 16 bit each, separated by colon(:) that is as follows.

xxxxxxxx:xxxxxxxx:xxxxxxxx:xxxxxxxx:xxxxxxxx:xxxxxxxx:xxxxxxxx

- Where each x is either 1 or 0. These 128 bits are then converted from binary to 8 sets of 4 hexadecimal digits each. Represented as follows.

xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx

- A typical IPv6 address would look like as follows.

2001:0db8:0000:0000:0000:8a2e:0370:7334

- Compared to IPv4, which can provide 2^{32} addresses, IPv6 can provide 2^{128} IP addresses. This equals approximately 340,282,366,920,938,463,463,374,607,431,768,211,456 possible addresses for devices, networks, internets, and intranets. However, the actual number of usable addresses is slightly lower because some address blocks are reserved for special purposes.
- In addition to offering a huge address space, IPv6 provides several advantages over IPv4. These include efficient route aggregation, which helps reduce the size of routing tables in routers, minimized packet header processing for faster routing, support for multicasting, Stateless Address Auto Configuration (SLAAC), improved security, and better overall network performance.
- The transition from IPv4 to IPv6 is challenging because the two protocols are not directly interoperable due to differences in their packet headers. As a result, direct communication between IPv4 and IPv6 devices is not possible without special transition mechanisms. Nevertheless, the migration from IPv4 to IPv6 is currently in progress worldwide.

IPv6 Packet Format

Internet Protocol Version 6 packet format defines how data is organized and transmitted in an IPv6 network.

An IPv6 packet mainly contains:

1. **IPv6 Header**
2. **Extension Header (optional)**
3. **Payload/Data**

IPv6 Header Format

The IPv6 header has a **fixed size of 40 bytes**.

Field	Size	Description
Version	4 bits	Specifies IP version number (6)
Traffic Class	8 bits	Controls packet priority and QoS
Flow Label	20 bits	Identifies packets belonging to the same flow
Payload Length	16 bits	Indicates size of payload data
Next Header	8 bits	Specifies the type of next header/protocol
Hop Limit	8 bits	Limits the number of hops/routers
Source Address	128 bits	IPv6 address of sender
Destination Address	128 bits	IPv6 address of receiver

Version (4 bits)

- Indicates the IP version.
- For IPv6, the value is always **6**.

Traffic Class (8 bits)

- Used for packet priority and traffic management.
- Helps in Quality of Service (QoS).

Example:

- Video call packets get higher priority than email packets.

Flow Label (20 bits)

- Identifies packets belonging to the same communication flow.
- Useful in real-time applications like:
 - Video streaming
 - Online gaming
 - Voice calls

Payload Length (16 bits)

- Specifies the size of payload data in bytes.
- Does not include the main IPv6 header size.

Next Header (8 bits)

- Indicates what comes after the IPv6 header.
- It may point to:
 - TCP
 - UDP
 - ICMPv6
 - Extension header

Example:

- Value 6 → TCP
- Value 17 → UDP

Hop Limit (8 bits)

- Similar to TTL in IPv4.
- Decreases by 1 at every router.
- Packet is discarded when it becomes 0.

Purpose:

- Prevents infinite looping of packets.

Source Address (128 bits)

- Contains sender's IPv6 address.

Example:

2001:db8::1

Destination Address (128 bits)

- Contains receiver's IPv6 address.

Example:

2001:db8::2

Extension Headers in IPv6

IPv6 uses optional extension headers for extra functions.

Examples:

- Routing header
- Fragmentation header
- Authentication header
- Security header

These are placed between the main header and payload.

Advantages of IPv6 Packet Format

Advantage	Description
Simplified Header	Faster processing
Fixed Header Size	Easier routing
Better QoS	Traffic class and flow label support
Improved Security	IPsec support
Efficient Routing	Reduced overhead

Difference Between IPv4 and IPv6 Packet Header

IPv4	IPv6
Variable header size	Fixed 40-byte header
Has checksum	No checksum
Complex header	Simplified header
32-bit addresses	128-bit addresses
Fragmentation by routers	Fragmentation by sender only

Why is IPv6 Header a Simplified Header?

IPv6 Header is called a **simplified header** because its header structure is simpler and more efficient than the IPv4 header.

Reasons for Simplification

Reason	Explanation
Fixed Header Size	IPv6 header size is always 40 bytes , making processing easier.
Fewer Fields	Some unnecessary IPv4 fields are removed.
No Header Checksum	Routers do not need to calculate checksum repeatedly, increasing speed.
Fragmentation Removed from Routers	Only sender performs fragmentation, reducing router workload.
Options Moved to Extension Headers	Optional information is separated from the main header.
Faster Routing	Simpler structure helps routers forward packets quickly.

Transition Mechanisms:

Dual Stack and Tunneling in IPv6 Transition

During the transition from Internet Protocol Version 4 to Internet Protocol Version 6, two important techniques are used:

1. **Dual Stack**
2. **Tunneling**

1. Dual Stack

Definition

Dual Stack is a transition mechanism where a device or network runs both IPv4 and IPv6 simultaneously.

That means:

- The device has:
 - an IPv4 address
 - and an IPv6 address

It can communicate using either protocol.

Working of Dual Stack

Device



- If destination supports IPv6 → IPv6 is used
- Otherwise → IPv4 is used

Advantages

- Smooth transition from IPv4 to IPv6
- Supports both old and new networks
- No major network interruption

Disadvantages

- Requires more resources
- Network management becomes complex

Example

A computer may have:

- IPv4: 192.168.1.10
- IPv6: 2001:db8::10

2. Tunneling

Definition

Tunneling is a technique where IPv6 packets are encapsulated inside IPv4 packets to travel through an IPv4 network.

Used when:

- Two IPv6 networks are separated by an IPv4 network.

Working of Tunneling

IPv6 Network → IPv4 Network → IPv6 Network
(IPv6 packet inside IPv4 packet)

Steps:

1. IPv6 packet is created
2. It is encapsulated inside an IPv4 packet
3. Travels through IPv4 network
4. IPv4 header is removed at destination
5. Original IPv6 packet is delivered

Advantages

- Allows IPv6 communication over IPv4 infrastructure
- No need to replace entire network immediately

Disadvantages

- Extra processing due to encapsulation
- Can reduce performance slightly

Difference Between Dual Stack and Tunneling

Dual Stack	Tunneling
Runs IPv4 and IPv6 together	Carries IPv6 through IPv4 network
Device supports both protocols	IPv6 packet encapsulated in IPv4
No encapsulation needed	Encapsulation required
Simpler communication	Used for compatibility across IPv4 networks

Translation Techniques:

Translation Techniques in IPv6 Transition

Internet Protocol Version 4 and Internet Protocol Version 6 are not directly compatible. So, **translation techniques** are used to enable communication between IPv4 and IPv6 devices.

What is Translation?

Translation is the process of converting:

- IPv4 packets ↔ IPv6 packets

so that devices using different IP versions can communicate.

Common Translation Techniques

Technique Function

NAT64 Allows IPv6 clients to communicate with IPv4 servers

DNS64 Generates IPv6 addresses from IPv4 addresses

SIIT Stateless IP/ICMP translation

NAT-PT Converts IPv4 packets to IPv6 packets and vice versa

1. NAT64

Definition

NAT64 (Network Address Translation 64) enables communication between IPv6-only clients and IPv4-only servers.

Working

IPv6 Client → NAT64 Gateway → IPv4 Server

- NAT64 converts:
 - IPv6 packets → IPv4 packets
 - IPv4 packets → IPv6 packets

Advantage

- IPv6 users can access IPv4 websites/services.

2. DNS64

Definition

DNS64 works with NAT64 by creating a fake IPv6 address from an IPv4 address.

Working

IPv4 Address → DNS64 → Generated IPv6 Address

Example:

- IPv4: 192.0.2.1
- DNS64 generates corresponding IPv6 format.

3. SIIT

Full Form

Stateless IP/ICMP Translation

Definition

Converts IPv4 headers into IPv6 headers without keeping connection state information.

Feature

- Fast translation
- No session tracking required

4. NAT-PT

Full Form

Network Address Translation – Protocol Translation

Function

Translates:

- IPv4 addresses ↔ IPv6 addresses
- IPv4 headers ↔ IPv6 headers

Limitation

- Complex
- Less commonly used now